

Mixed Partial Derivatives: a detailed solved example

AS1109 Calculus and Ordinary Differential Equations – Section H

Supplement to Lesson 5: higher-order partial derivatives and Euler's / Mixed Derivative Theorem

Main example. Consider

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0), \end{cases}$$

and show carefully that the mixed partial derivatives at the origin exist, but

$$f_{xy}(0, 0) = -1 \neq 1 = f_{yx}(0, 0).$$

So this function gives a counterexample to the careless statement “mixed partials are always equal”. The correct theorem needs extra hypotheses.

1. What should you already know?

Before this example, you should be comfortable with the following basic facts.

1. The partial derivative definitions at a point:

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}, \quad f_y(a, b) = \lim_{k \rightarrow 0} \frac{f(a, b + k) - f(a, b)}{k}.$$

2. Away from problematic points, you may often compute f_x and f_y by treating the other variable as a constant.
3. The product rule:

$$(uv)' = u'v + uv'.$$

When differentiating partially, apply the same rule while holding the other independent variable fixed.

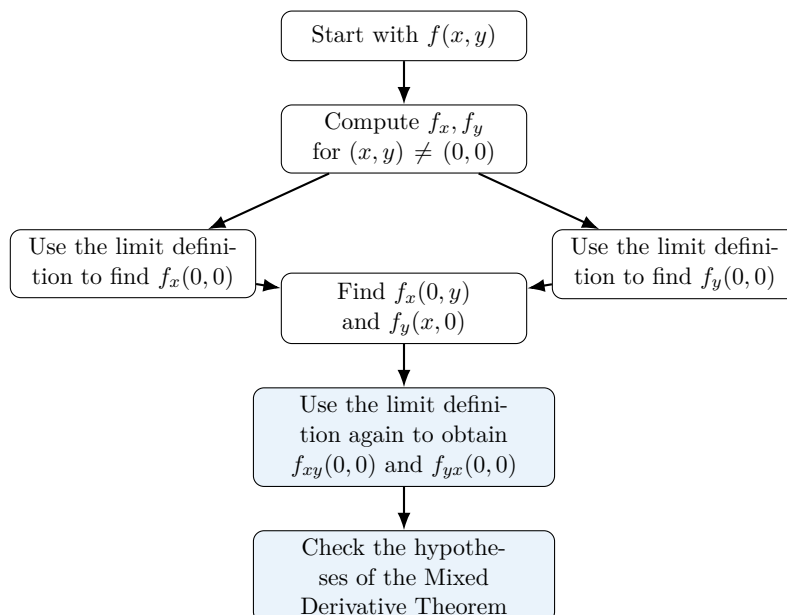
4. The quotient rule:

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2} \quad (v \neq 0).$$

Again, in a partial derivative the other independent variable is treated as a constant.

5. For the mixed derivatives,

$$f_{xy}(a, b) = \frac{\partial}{\partial y}(f_x(a, b)), \quad f_{yx}(a, b) = \frac{\partial}{\partial x}(f_y(a, b)).$$



2. Step 0: is the function even continuous at the origin?

For this piecewise definition,

$$f : \mathbb{R}^2 \longrightarrow \mathbb{R}.$$

Thus the **domain** is all of $\mathbb{R}^2$; the special definition $f(0, 0) = 0$ fills in the point where the rational formula alone would be undefined. The **codomain** is $\mathbb{R}$. In fact the **range is also** $\mathbb{R}$: in polar form,

$$f(r, \theta) = \frac{r^2}{4} \sin(4\theta),$$

and by choosing an angle with $\sin(4\theta) = 1$ or -1 and then varying $r \geq 0$, every real value can be obtained.

We first check continuity at the origin. For $(x, y) \neq (0, 0)$,

$$|f(x, y)| = \left| \frac{xy(x^2 - y^2)}{x^2 + y^2} \right|.$$

Now

$$|x^2 - y^2| \leq x^2 + y^2,$$

so

$$|f(x, y)| \leq |xy|.$$

Since $|xy| \rightarrow 0$ as $(x, y) \rightarrow (0, 0)$, it follows by squeezing that

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0 = f(0, 0).$$

Therefore f is **continuous at** $(0, 0)$.

Continuity of f itself is *not enough* for Euler's / Mixed Derivative Theorem. The theorem needs conditions on the *second partial derivatives*.

3. Step 1: compute f_x and f_y for $(x, y) \neq (0, 0)$

For $(x, y) \neq (0, 0)$, write

$$f(x, y) = \frac{N(x, y)}{D(x, y)}, \quad N(x, y) = x^3y - xy^3, \quad D(x, y) = x^2 + y^2.$$

3.1 Computing f_x

Treat y as a constant. Then

$$N_x = 3x^2y - y^3, \quad D_x = 2x.$$

Using the quotient rule,

$$f_x = \frac{N_x D - N D_x}{D^2}.$$

Therefore

$$f_x(x, y) = \frac{(3x^2y - y^3)(x^2 + y^2) - (x^3y - xy^3)(2x)}{(x^2 + y^2)^2}.$$

Now simplify the numerator:

$$\begin{aligned} & (3x^2y - y^3)(x^2 + y^2) - (x^3y - xy^3)(2x) \\ &= 3x^4y + 3x^2y^3 - x^2y^3 - y^5 - 2x^4y + 2x^2y^3 \\ &= x^4y + 4x^2y^3 - y^5 \\ &= y(x^4 + 4x^2y^2 - y^4). \end{aligned}$$

Hence

$$\boxed{f_x(x, y) = \frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^2}}, \quad (x, y) \neq (0, 0).$$

3.2 Computing f_y

Treat x as a constant. Then

$$N_y = x^3 - 3xy^2, \quad D_y = 2y.$$

Again,

$$f_y = \frac{N_y D - N D_y}{D^2}.$$

Thus

$$f_y(x, y) = \frac{(x^3 - 3xy^2)(x^2 + y^2) - (x^3y - xy^3)(2y)}{(x^2 + y^2)^2}.$$

Simplifying the numerator,

$$\begin{aligned} & (x^3 - 3xy^2)(x^2 + y^2) - (x^3y - xy^3)(2y) \\ &= x^5 + x^3y^2 - 3x^3y^2 - 3xy^4 - 2x^3y^2 + 2xy^4 \\ &= x^5 - 4x^3y^2 - xy^4 \\ &= x(x^4 - 4x^2y^2 - y^4). \end{aligned}$$

So

$$\boxed{f_y(x, y) = \frac{x(x^4 - 4x^2y^2 - y^4)}{(x^2 + y^2)^2}}, \quad (x, y) \neq (0, 0).$$

4. Step 2: compute $f_x(0, 0)$ and $f_y(0, 0)$ from the limit definitions

At the origin, we *must* use the definition.

4.1 Computing $f_x(0, 0)$

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h}.$$

But $f(h, 0) = 0$ because the numerator contains the factor xy , and $f(0, 0) = 0$. Hence

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0.$$

4.2 Computing $f_y(0,0)$

Similarly,

$$f_y(0,0) = \lim_{k \rightarrow 0} \frac{f(0,k) - f(0,0)}{k} = 0.$$

So the first partial derivatives exist at the origin and

$$\boxed{f_x(0,0) = 0, \quad f_y(0,0) = 0.}$$

Method note. At ordinary points we often differentiate by the usual rules. At a special point such as $(0,0)$ for a piecewise-defined function, use the limit definition unless a previously established derivative formula is valid there.

5. Step 3: prepare for the mixed derivatives

To compute $f_{xy}(0,0)$, we need $f_x(0,y)$. To compute $f_{yx}(0,0)$, we need $f_y(x,0)$.

Using the formula for f_x with $x = 0$ and $y \neq 0$,

$$f_x(0,y) = \frac{y(0 + 0 - y^4)}{(0 + y^2)^2} = -y.$$

Also $f_x(0,0) = 0$. Hence

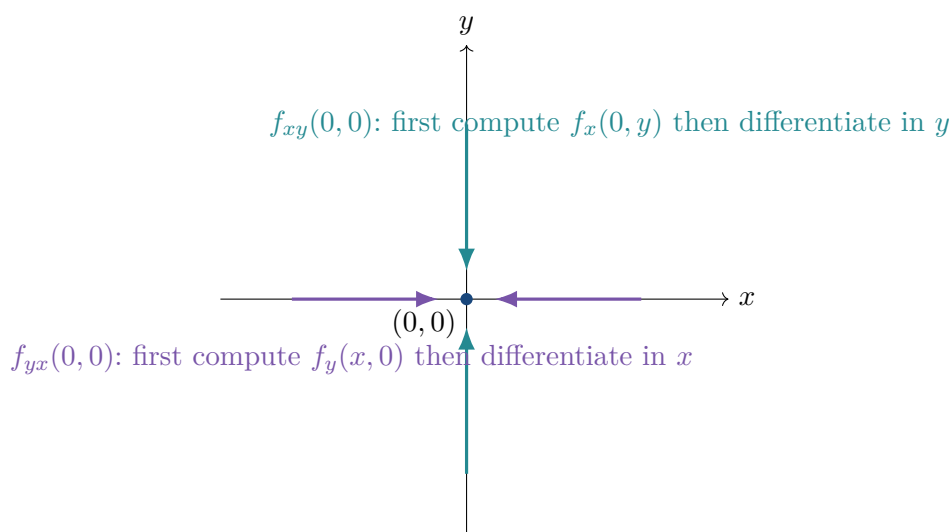
$$\boxed{f_x(0,y) = \begin{cases} -y, & y \neq 0, \\ 0, & y = 0. \end{cases}}$$

Similarly, using the formula for f_y with $y = 0$ and $x \neq 0$,

$$f_y(x,0) = \frac{x(x^4 - 0 - 0)}{(x^2 + 0)^2} = x,$$

and $f_y(0,0) = 0$. Thus

$$\boxed{f_y(x,0) = \begin{cases} x, & x \neq 0, \\ 0, & x = 0. \end{cases}}$$



6. Step 4: compute the mixed partials at the origin

6.1 Computing $f_{xy}(0, 0)$

By definition,

$$f_{xy}(0, 0) = \lim_{k \rightarrow 0} \frac{f_x(0, k) - f_x(0, 0)}{k}.$$

From the previous step, $f_x(0, k) = -k$ for $k \neq 0$ and $f_x(0, 0) = 0$. Therefore

$$f_{xy}(0, 0) = \lim_{k \rightarrow 0} \frac{-k - 0}{k} = -1.$$

So

$$\boxed{f_{xy}(0, 0) = -1.}$$

6.2 Computing $f_{yx}(0, 0)$

Similarly,

$$f_{yx}(0, 0) = \lim_{h \rightarrow 0} \frac{f_y(h, 0) - f_y(0, 0)}{h}.$$

Now $f_y(h, 0) = h$ and $f_y(0, 0) = 0$. So

$$f_{yx}(0, 0) = \lim_{h \rightarrow 0} \frac{h - 0}{h} = 1.$$

Hence

$$\boxed{f_{yx}(0, 0) = 1.}$$

Conclusion.

$$\boxed{f_{xy}(0, 0) = -1 \neq 1 = f_{yx}(0, 0).}$$

Therefore this function does **not** satisfy the conclusion of Euler's / Mixed Derivative Theorem at the origin.

7. Step 5: why does the theorem fail here?

The standard theorem says, roughly:

If f_x , f_y , f_{xy} and f_{yx} are defined in an open region containing (a, b) and the mixed partial derivatives are continuous at (a, b) (or, in many textbooks, continuous on a neighbourhood of (a, b)), then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

So to understand our example, we should check continuity of the mixed partials at the origin.

7.1 Checking f_{xy} near $(0, 0)$

Take $x \neq 0$ and set $y = 0$. Since $f_x(x, 0) = 0$, we have

$$f_{xy}(x, 0) = \lim_{k \rightarrow 0} \frac{f_x(x, k) - f_x(x, 0)}{k} = \lim_{k \rightarrow 0} \frac{f_x(x, k)}{k}.$$

Using the formula for f_x ,

$$\frac{f_x(x, k)}{k} = \frac{x^4 + 4x^2k^2 - k^4}{(x^2 + k^2)^2}.$$

Letting $k \rightarrow 0$ gives

$$f_{xy}(x, 0) = 1 \quad (x \neq 0).$$

But at the origin we already found $f_{xy}(0, 0) = -1$. Therefore f_{xy} is **not continuous** at $(0, 0)$.

7.2 Checking f_{yx} near $(0, 0)$

Now take $y \neq 0$ and set $x = 0$. Since $f_y(0, y) = 0$, we get

$$f_{yx}(0, y) = \lim_{h \rightarrow 0} \frac{f_y(h, y) - f_y(0, y)}{h} = \lim_{h \rightarrow 0} \frac{f_y(h, y)}{h}.$$

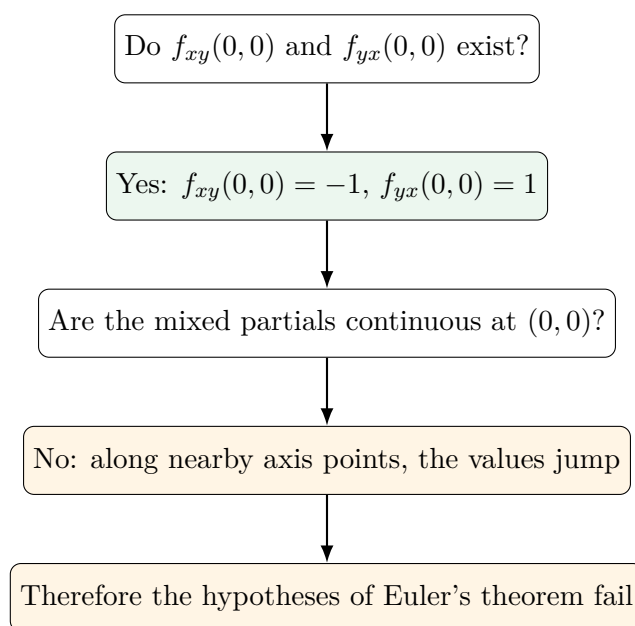
Using the formula for f_y ,

$$\frac{f_y(h, y)}{h} = \frac{h^4 - 4h^2y^2 - y^4}{(h^2 + y^2)^2}.$$

Letting $h \rightarrow 0$ gives

$$f_{yx}(0, y) = -1 \quad (y \neq 0).$$

But at the origin we found $f_{yx}(0, 0) = 1$. Hence f_{yx} is also **not continuous at** $(0, 0)$.



8. A quick summary table

Quantity	Value / conclusion
$f(0, 0)$	0
f at $(0, 0)$	continuous
$f_x(0, 0)$ and $f_y(0, 0)$	both exist and equal 0
$f_x(x, y)$ for $(x, y) \neq (0, 0)$	$\frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^2}$
$f_y(x, y)$ for $(x, y) \neq (0, 0)$	$\frac{x(x^4 - 4x^2y^2 - y^4)}{(x^2 + y^2)^2}$
$f_x(0, y)$	$-y$
$f_y(x, 0)$	x
$f_{xy}(0, 0)$	-1
$f_{yx}(0, 0)$	1
Are f_{xy} and f_{yx} equal at $(0, 0)$?	No
Why does Euler's theorem not apply?	The mixed partials are not continuous at $(0, 0)$; the theorem's hypotheses fail.

9. Summary of the method

1. For a piecewise-defined function at a special point, **always be ready to use the limit definition**.
2. The notation f_{xy} means: differentiate first with respect to x , then with respect to y .
3. Mixed partials are often equal for nice functions, but not always.
4. In theorem questions, **never forget the hypotheses**. The conclusion

$$f_{xy}(a, b) = f_{yx}(a, b)$$

needs continuity assumptions on the mixed partials.

5. In this example, the function itself is continuous at the origin, and the first partials exist there, but the mixed partial derivatives fail to be continuous there. That is exactly why the theorem breaks down.

Conceptual summary. This example combines the definitions of f_x , f_y , f_{xy} and f_{yx} with ordinary differentiation rules and the hypotheses of the Mixed Derivative Theorem. It shows that existence of first and mixed partial derivatives at a point does not by itself guarantee equality of the mixed partials.